PowerPoint Presentation by

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MANAGEMENT ACCOUNTING

8th EDITION

BY

HANSEN & MOWEN

11 COST-VOLUME-PROFIT ANALYSIS

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

LEARNING OBJECTIVES

1. Determine the number of units sold to break even or earn a targeted profit.
2. Calculate the amount of revenue required to break even or earn a targeted profit.
3. Apply cost-volume-profit analysis in a multiple-product setting.

Continued

LEARNING OBJECTIVES

4. Prepare a profit-volume graph & a cost-volume-profit graph, and explain the meaning of each.
5. Explain the impact of risk, uncertainty, & changing variables on cost-volume-profit analysis.
6. Discuss the impact of activity-based costing on cost-volume-profit analysis.

Click the button to skip Questions to Think About

QUESTIONS TO THINK ABOUT:
McFarland's Fine Foods

What kinds of variable & fixed costs do you think Janet will incur?

QUESTIONS TO THINK ABOUT:
McFarland's Fine Foods

Given Bob's initial assessment that variable costs are higher than the price, what is wrong with Janet's though that selling more is the way to go?

QUESTIONS TO THINK ABOUT:
McFarland's Fine Foods

How important is break-even analysis to a firm? Do you suppose that large companies do break-even analysis as well as small companies?

QUESTIONS TO THINK ABOUT:
McFarland's Fine Foods

Why is the concept of breaking even important? Doesn't Janet want to make a profit? If Janet doesn't know what price to charge, how could she get a better idea?

LEARNING OBJECTIVE

1

Determine the number of units sold to break even or earn a targeted profit.

COST-VOLUME-PROFIT (CVP)

CVP expresses:

- # units that must be sold to break even
- Impact of a given reduction in fixed costs on break-even point
- Impact of an increase in price on profit
- Sensitivity analysis of impact of various price or cost levels on profit

BREAK-EVEN POINT: Definition

Is the point where total revenue equals total cost; the point of zero profit.

UNIT: Examples

- Examples of unit of measurement
  - Procter & Gamble: bar of Ivory soap
  - McFarland: jar of salsa
  - Southwest Air Lines
    - Passenger mile
    - One-way trip

FORMULA: Operating Income

Operating income includes revenues & expenses from the firm's normal operations.

Operating Income

= Sales revenue

– Variable expenses

– Fixed expenses

NET INCOME: Definition

Is operating income minus income taxes.

WHITTIER CO.: Background

Operating income for mulching lawn mower

Sales (1,000 units @ $400)	$ 400,000
Less: Variable expenses	325,000
Contribution margin	$ 75,000
Less: Fixed expenses	45,000
Operating income	$ 30,000

FORMULA: Break-Even

Break-even is 0 profit.

Break-even:

0 = Sales revenue – Variable expenses – Fixed expenses

0 = ($400 *x* Units) – ($325 *x* Units) - $45,000

($75 *x* Units) = $45,000

Units = 600

WHITTIER CO.: √Income Statement

√Check-up on break-even

Sales (600 units @ $400)	$ 240,000
Less: Variable expenses	195,000
Contribution margin	$ 45,000
Less: Fixed expenses	45,000
Operating income	**$ 0**

CONTRIBUTION MARGIN:
Definition

Is sales revenue minus variable costs (Sales – VC).

FORMULA: Break-Even

Break-even using contribution margin.

Break-even units:

 # Units = Fixed cost / Unit contribution margin

 # Units = $45,000 / ($400 - $325)

= 600

WHITTIER CO.: √Income Statement

√Check-up on break-even

Sales (600 units @ $400)	$ 240,000
Less: Variable expenses	195,000
Contribution margin	$ 45,000
Less: Fixed expenses	45,000
Operating income	**$ 0**

FORMULA: Target Profit

Target profit is profit desired.

Target profit in dollars:

\$ 60,000 = (\$400 *x* Units) – (\$325 *x* Units) - \$45,000

\$105,000 = \$75,000 *x* Units

Units = 1,400

FORMULA: Target Profit in Units

Target profit is profit desired.

Target profit in units:

$$\# \text{ Units} = \frac{(\text{Fixed cost} + \text{Target profit})}{\text{Unit contribution margin}}$$

 # Units = ($45,000 + $60,000) / ($400 - $325)

 # Units = 1,400

WHITTIER CO.: √Income Statement

√Check-up on target profit

Sales (600 units @ $400)	$ 560,000
Less: Variable expenses	455,000
Contribution margin	$ 105,000
Less: Fixed expenses	45,000
Operating income	**$ 60,000**

Can we calculate target profit by another way than dollars & units?

Yes! Target income can be calculated as a **percent of revenue**.

FORMULA: Target Profit % Sales

Target profit can be calculated as % of revenue.

Target profit as % of sales:

<u>0.15</u> (\$400 x Units) =

(\$400 *x* Units) – (\$325 *x* Units) - \$45,000

\$60 *x* Units = (\$75 x Units) - \$45,000

 # Units = 3,000

Can we calculate units required to produce after tax target profits?

Yes! First, calculate the level of operating profit, then translate into number of units.

FORMULA: After-Tax Target Profit

If Whittier has a 35% tax rate & wants Net income (after-tax profit) of $48,750.

After-tax target profit:

Net income = Operating income (1 – Tax rate)

$48,750 = Operating income (1 – 0.35)

$75,000 = Operating income

WHITTIER CO.: √Income Statement

√Check-up on target profit

Sales (1,600 units @ $400)	$ 640,000
Less: Variable expenses	520,000
Contribution margin	$ 120,000
Less: Fixed expenses	45,000
Operating income	$75,000
Less: Income taxes (35%)	26,250
Net income	**$ 48,750**

LEARNING OBJECTIVE

2

Calculate the amount of revenue required to break even or earn a targeted profit.

VARIABLE COST RATIO:
Definition

Is the proportion of each sales dollar used to cover variable costs.

CONTRIBUTION MARGIN RATIO:
Definition

Is the proportion of each sales dollar available to cover fixed costs & provide profit.

Where do fixed costs fit in?

Fixed costs can equal CM (break-even), or be less than CM (profit) or more than CM (loss).

WHITTIER CO.: Background

CMR for mulching lawn mower.

Sales (1,000 units @ $400)	$ 400,000	100.00%
Less: Variable expenses	325,000	81.25%
Contribution margin	$ 75,000	**18.75%**
Less: Fixed expenses	45,000	
Operating income	$ 30,000	

FORMULA: Break-Even CMR

Contribution margin ratio (CMR) makes calculation easier.

0 = Sales (1 – VC rate) – Fixed Costs

= Sales (1 – 0.8125) - $45,000

Sales = $240,000

OR

Break-even Sales = Fixed cost / CMR

$240,000 = $45,000 / 0.1875

LEARNING OBJECTIVE

3

Apply cost-volume-profit analysis in a multiple-product setting.

Can we use CVP if Whittier has more than 1 product?

Yes. But we have to add direct fixed expenses into the analysis.

DIRECT FIXED EXPENSES:
Definition

Are fixed costs that can be traced to each product and would be avoided if the product did not exist.

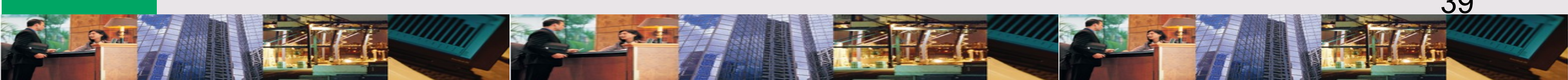
WHITTIER CO.: Sales Background

Operating income for multiple products.

	Mulching	**Riding**	**Total**
Sales (1,000 units @ $400)	$ 480,000	$640,000	$1,120,000
Less: Variable expenses	390,000	480,000	870,000
Contribution margin	$ 90,000	$160,000	$ 250,000
Less: Direct fixed exp.	**30,000**	**40,000**	**70,000**
Product margin	$ 60,000	$120,000	$ 180,000
Less: Fixed expenses			26,250
Operating income			$ 153,750

SALES MIX: Definition

Is the relative combination of products being sold.



WHITTIER CO.: Sales Mix & CVP Background

Margin for multiple products

Product	Unit Price	VC	CM	Package Cont. Mix	Margin*
Mulching	$400	$325	$ 75	3	$ 225
Riding	800	600	200	2	400
Package Total					$ 625

*Margin = Units in package *x* CM

FORMULA: Break-Even Multiple Products

If Whittier has 2 products, calculate break-even separately.

Break-Even = Fixed costs / (Price – Unit VC)

Mulching mower = $30,000 / $75

= 400 units

Riding mower = $40,000 / $200

= 200 units

FORMULA: Break-Even Packages

Contribution margin approach to multiple products.

Break-even packages = Fixed cost / Package CM

= $96,250 / $625

= 154 Packages

BREAK-EVEN SOLUTION

Mulching mower sales = $400 *x* 3 *x* 154 packages.

	Mulching Mower	Riding Mower	Total
Sales	$184,800	$246,400	$431,200
Less: Variable expenses	150,150	184,800	334,950
Contribution margin	$ 34,650	$ 61,600	$ 96,250
Less: Direct fixed expenses	30,000	40,000	70,000
Segment margin	$ 4,650	$ 21,600	$ 26,250
Less: Common fixed expenses			26,250
Operating income			$ 0

Exhibit 11-1 Income Statement: Break-Even Solution

EXHIBIT 11-1

WHITTIER CO.: CMR Background

Contribution margin ratio for multiple products

	Total	Percent
Sales (1,000 units @ $400)	$1,120,000	100.00%
Less: Variable expenses	870,000	.7767
Contribution margin	$ 250,000	**.2232**
Less: Direct fixed exp.	70,000	
Product margin	$ 180,000	
Less: Fixed expenses	26,250	
Operating income	$ 153,750	

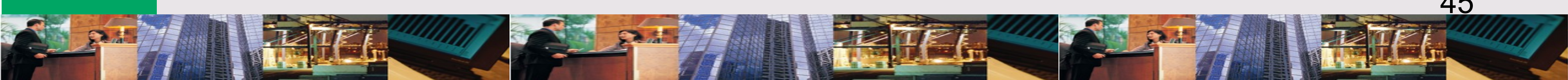
FORMULA: Break-Even Packages

Contribution margin approach to break-even for multiple products.

Break-even sales = Fixed cost / Package CMR

= $96,250 / 0.2232

= $431,228



LEARNING OBJECTIVE

4

Prepare a profit-volume graph & a cost-volume-profit graph, and explain the meaning of each.

PROFIT-VOLUME GRAPH

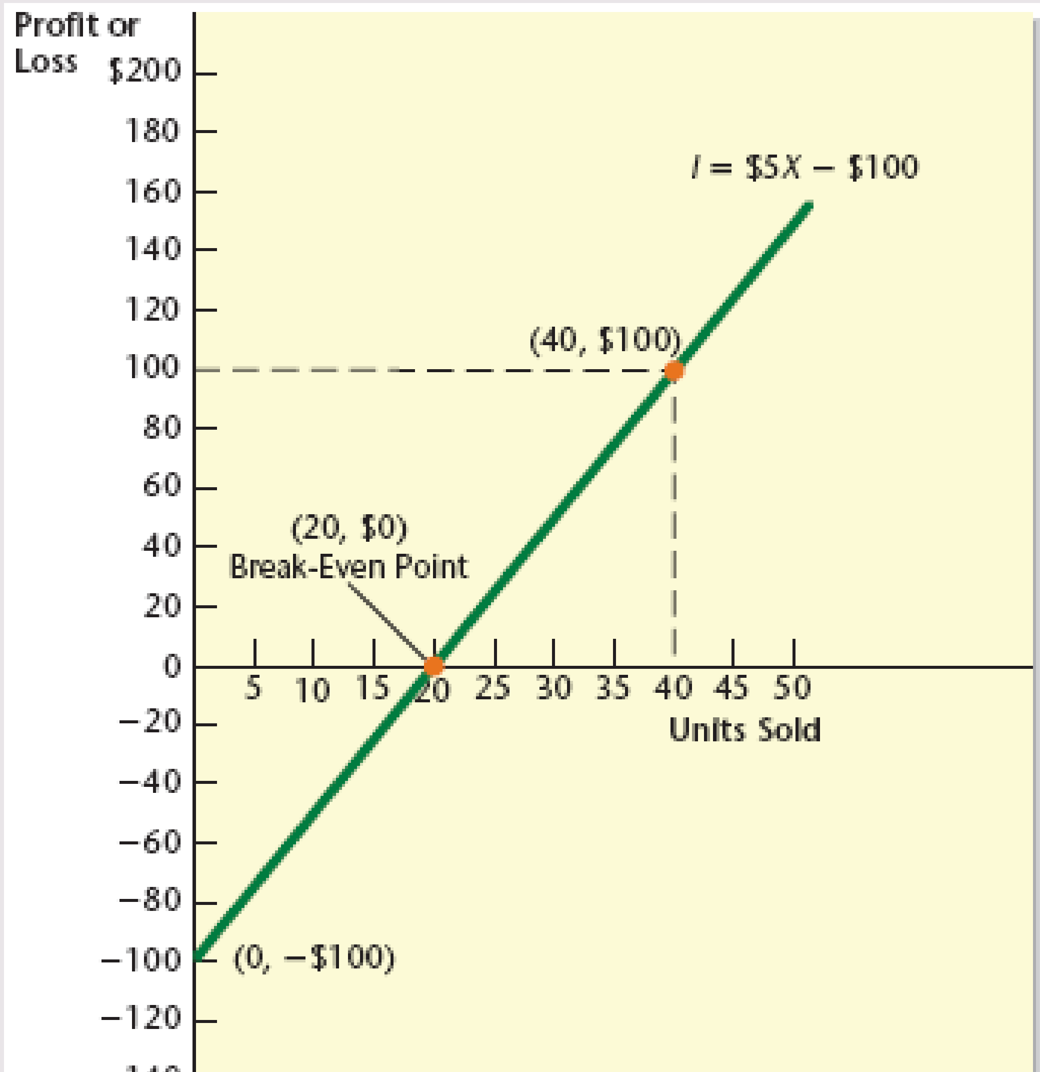


EXHIBIT 11-2

What happens when cost is added to a profit-volume graph?

The new graph clearly shows areas of profit & loss.

COST-PROFIT-VOLUME GRAPH

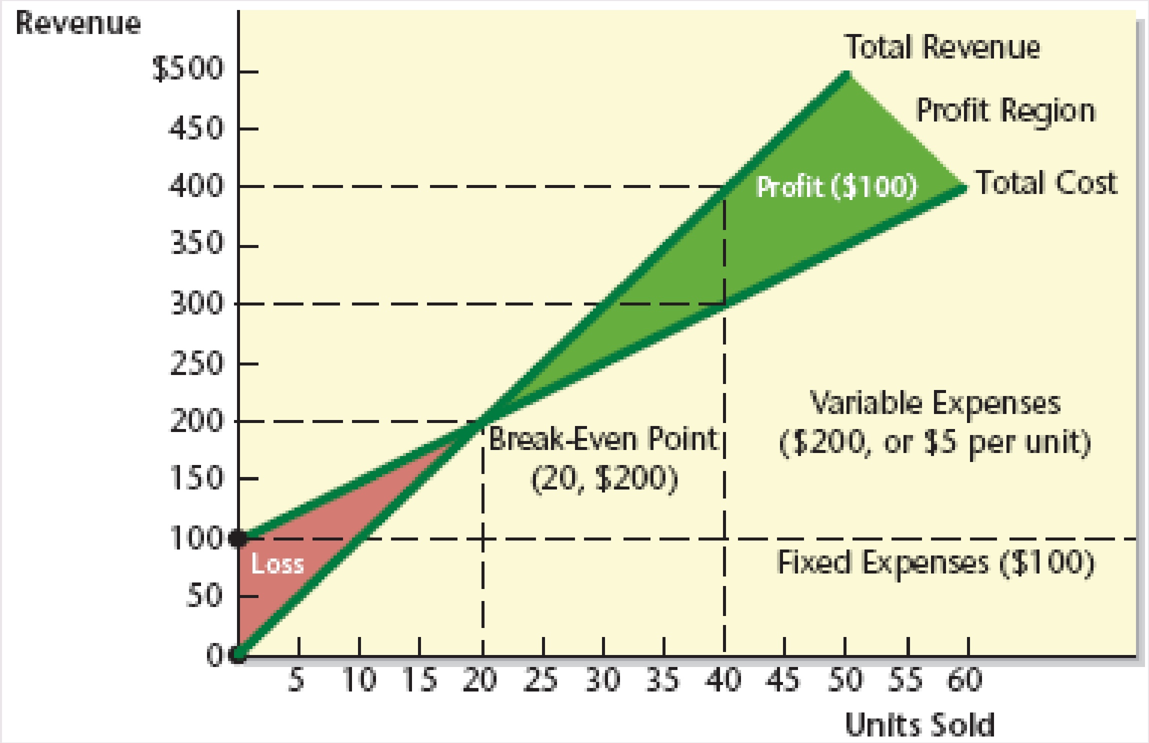


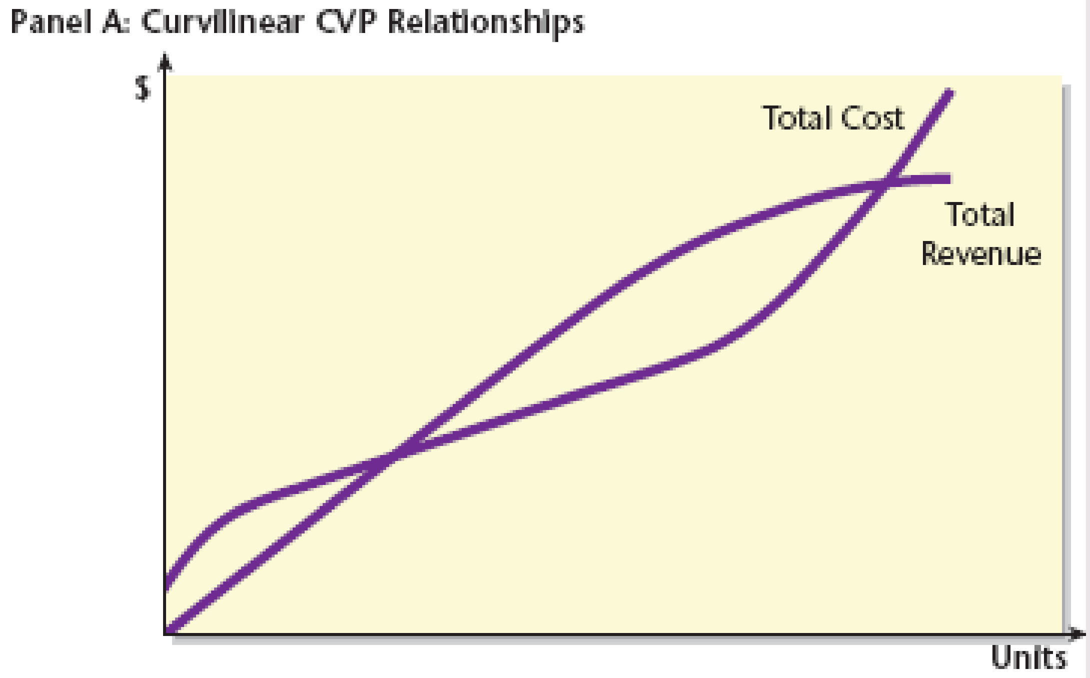
EXHIBIT 11-3

ASSUMPTIONS OF CVP

CVP analysis assumes

- Linear revenue & cost functions
- Price, total fixed costs, & unit variable costs can be accurately identified & remain constant over the relevant range
- What is produced is sold
- Sales mix is known
- Selling prices & costs are known with certainty

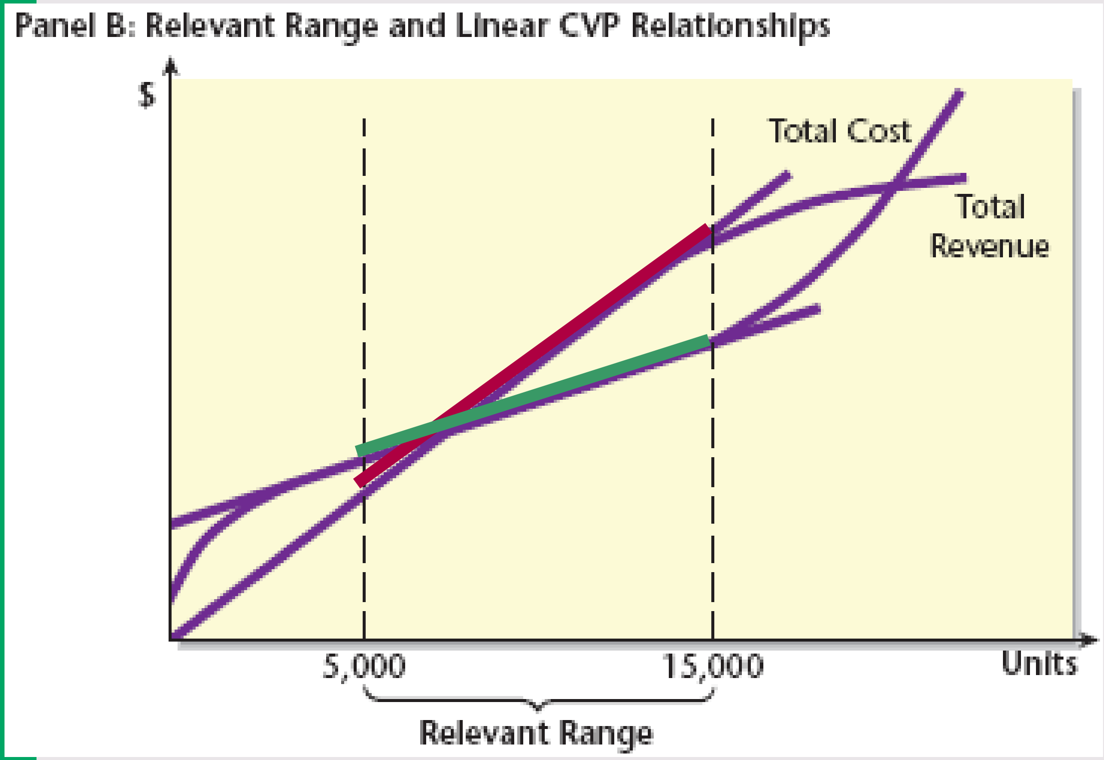
COST-PROFIT-VOLUME GRAPH



Will CVP apply if the cost & revenue functions are not linear?

EXHIBIT 11-4a

COST-PROFIT-VOLUME GRAPH



Yes. CVP will apply so long as the cost & revenue functions are approximately linear over the **relevant range**.

EXHIBIT 11-4b

LEARNING OBJECTIVE

5

Explain the impact of risk, uncertainty, & changing variables on cost-volume-profit analysis.

What can Whittier do to increase sales of mulching mowers?

Sales can be increased by some combination of **increased advertising** and **decreased prices.**

ALTERNATIVES

For the mulching mower are:

- **#1** If advertising expenditures increase by $8,000, sales will increase from 1,600 to 1,725 mowers.
- **#2** A price decrease from $400 to $375 per mower will increase sales from 1,600 to 1,900.
- **#3** Decreasing price to $375 and increasing advertising expenditures by $8,000 will increase sales from 1,600 to 2,600 mowers.

ALTERNATIVE #1

	Before the Increased Advertising	With the Increased Advertising
Units sold	1,600	1,725
Unit contribution margin	× $75	× $75
Total contribution margin	$120,000	$129,375
Less: Fixed expenses	45,000	53,000
Profit	$ 75,000	$ 76,375

	Difference in Profit
Change in sales volume	125
Unit contribution margin	× $75
Change in contribution margin	$9,375
Less: Change in fixed expenses	8,000
Increase in profit	$1,375

Exhibit 11-5 Summary of the Effects of Alternative 1

Increasing advertising increases profit by **$1,325**.

EXHIBIT 11-5

ALTERNATIVE #2

	Before the Proposed Price Decrease	With the Proposed Price Decrease
Units sold	1,600	1,900
Unit contribution margin	× $75	× $50
Total contribution margin	$120,000	$95,000
Less: Fixed expenses	45,000	45,000
Profit	$ 75,000	$50,000

	Difference in Profit
Change in contribution margin ($95,000 − $120,000)	$(25,000)
Less: Change in fixed expenses	—
Decrease in profit	$(25,000)

Exhibit 11-6 Summary of the Effects of Alternative 2

Decreasing price decreases profit by **$25,000**.

EXHIBIT 11-6

ALTERNATIVE #3

	Before the Proposed Price and Advertising Changes	With the Proposed Price Decrease and Advertising Increase
Units sold	1,600	2,600
Unit contribution margin	× $75	× $50
Total contribution margin	$120,000	$130,000
Less: Fixed expenses	45,000	53,000
Profit	$ 75,000	$ 77,000

	Difference in Profit
Change in contribution margin ($130,000 − $120,000)	$10,000
Less: Change in fixed expenses ($53,000 − $45,000)	8,000
Increase in profit	$ 2,000

Exhibit 11-7 Summary of the Effects of Alternative 3

The combination of increasing advertising & decreasing price increases profit by **$2,000**.

EXHIBIT 11-7

How should Whittier use the results of analysis in the 3 alternatives?

Whittier should consider its choice in the context of **Risk & Uncertainty.**

RISK & UNCERTAINTY

For Whittier, risk includes the fact that prices and costs can not be predicted with certainty. Risk assumes that the distributions of the variables in question are known (i.e., we know how sales will react in response to changes in price or cost). Under uncertainty, these distributions are not known.

What should Whittier consider in addressing risk & uncertainty?

Whittier should consider its **margin of safety, operating leverage, and sensitivity analysis to CVP**.

MARGIN OF SAFETY: Definition

Is the difference between break-even volume or sales and expected volume or sales.

OPERATING LEVERAGE:
Definition

Is the use of fixed costs (e.g., automated system) to extract higher percentage changes in profits.

SENSITIVITY ANALYSIS:
Definition

Is the use of "what if" to examine the impact of changes in underlying assumptions on operating results.

SENSITIVITY ANALYSIS

Under 2 systems, the same change in price will have different effects on elements of CVP, & response to risk & uncertainty.

	Manual System	Automated System
Price	Same	Same
Variable cost	Relatively Higher	Relatively Lower
Fixed cost	Relatively Lower	Relatively Higher
Contribution margin	Relatively Lower	Relatively Higher
Break-even point	Relatively Lower	Relatively Higher
Margin of safety	Relatively Higher	Relatively Lower
Degree of operating leverage	Relatively Lower	Relatively Higher
Down-side risk	Relatively Lower	Relatively Higher
Up-side potential	Relatively Lower	Relatively Higher

Exhibit 11-8 Differences between Manual and Automated Systems

EXHIBIT 11-8

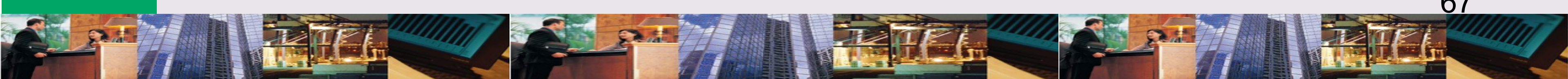
LEARNING OBJECTIVE

6

Discuss the impact of activity-based costing on cost-volume-profit analysis.

ABC & CVP

ABC divides costs into unit-based & non-unit-based categories. CVP has to adjust its formulas to incorporate this division.



FORMULA: ABC & CVP

ABC breaks CVP up into unit variable costs & other non-unit level costs.

Break-even =

[Fixed cost + (Unit VC x # Units)

+ (Setup cost x # Setups)

+ Engineering cost x # Engineering hours)] ÷

(Price – Unit variable cost)

ABC & CVP: Background

Data to compare conventional & ABC analysis.

Activity Driver	Unit VC	Level of Activity Driver
Units sold	$10	----
Setups	1,000	20
Engineering hours	30	1,000
Other data:		
Total fixed costs (conv.)		$ 100,000
Total fixed costs (ABC)		50,000
Unit selling price		20

FORMULA: Conventional CVP

 # Units

= [Targeted income + Conventional fixed cost] ÷

(Price – Unit variable cost)

= ($20,000 + $100,000) ÷ ($20 - $10)

= 12,000 Units

FORMULA: ABC CVP

\# Units

= [Targeted income + ABC Fixed cost +

(Setup cost x # Setups) +

(Engineering cost x # Engineering hours)] ÷

(Price – Unit variable cost)

= ($20,000 + $50,000 + ($1,000 x 20) + ($30 x 1,000)] ÷ ($20 - $10)

= 12,000 Units

What are the strategic implications of the 2 approaches to analyzing CVP?

An ABC approach to CVP allows for a **better defined breakdown** of costs to analyze alternative recommendations.

CHAPTER 11

THE END